

House Price Prediction

Week 1 Internship Project | Housing Prices Dataset (Kaggle)

Dataset Overview

The dataset contains **545 records** and **13 columns**. The target variable is **price (₹)**. Features include area (sq ft), bedrooms, bathrooms, stories, parking, and binary amenity flags (air conditioning, basement, main road access, preferred area, furnishing status). No missing values or duplicates were found. Categorical columns were one-hot encoded before modelling.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	Rs.215,052	Rs.270,851	0.8474
Random Forest	Rs.275,997	Rs.346,660	0.75

Linear Regression outperformed Random Forest ($R^2 = 0.847$ vs 0.750), explaining ~85% of price variance. On average, predictions were within **₹2.15 lakh** of actual prices — strong accuracy for a simple linear model with no hyperparameter tuning.

Which Features Influence Price the Most?

Area is the single strongest driver (importance ≈ 0.65) — far ahead of all other features. The next most influential are **bathrooms**, **bedrooms**, **preferred area**, and **stories**. Interestingly, each additional bathroom adds more value than an extra bedroom, and properties in a preferred area command a consistent premium regardless of other features.

What Was Surprising?

Hot water heating had a weaker correlation with price than expected, despite being a comfort feature. The price distribution was right-skewed with a mean of **₹3,065,466** — a small number of luxury properties pull the average significantly above the median.

Recommendation for a Real Estate Business

Prioritise **maximising usable area** and **adding bathrooms** before listing — these yield the highest price uplift per rupee invested. Targeting acquisition in **preferred-area localities** offers a reliable premium. Installing **air conditioning** is a cost-effective upgrade that measurably boosts listing value.